2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples.

CODING:

```
import pandas as pd


data = pd.DataFrame([
   ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
   ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
   ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
   ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
], columns=['Sky', 'Temp', 'Humidity', 'Wind',
         'Water', 'Forecast', 'EnjoySport'])


S = ['0'] * 6
G = [['?'] * 6]


for _, row in data.iterrows():
   x = row.iloc[:-1].tolist()


if row.iloc[-1] == 'Yes':
   for i in range(6):
      if S[i] == '0':
         S[i] = x[i]
      elif S[i] != x[i]:
         S[i] = '?'


   G = [g for g in G if all(
```

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```
        g[i] == '?' or g[i] == S[i] or S[i] == '?'
        for i in range(6)
    )]

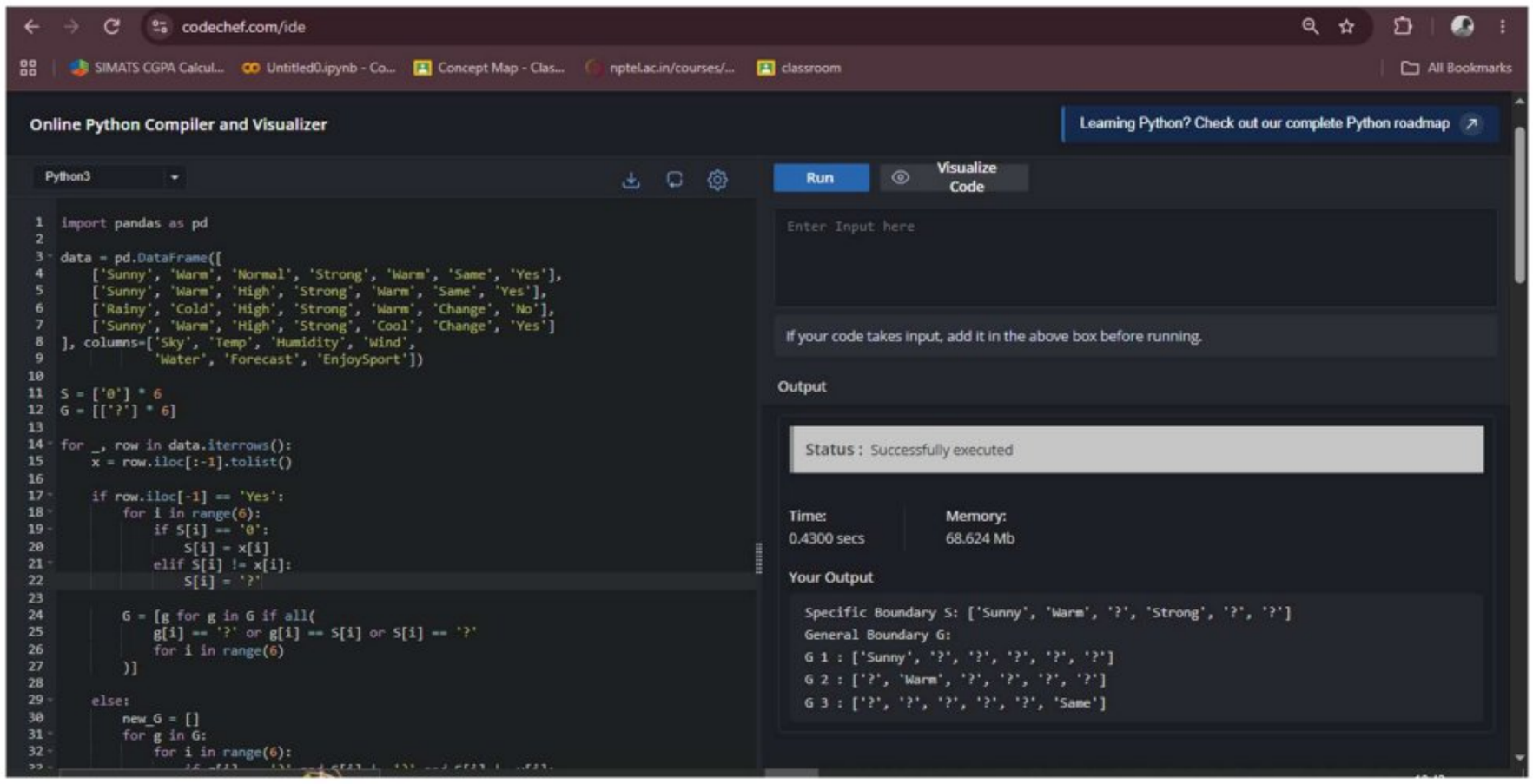
else:
    new_G = []
    for g in G:
        for i in range(6):
            if g[i] == '?' and S[i] != '?' and S[i] != x[i]:
                h = g.copy()
                h[i] = S[i]
                new_G.append(h)
        G = new_G if new_G else G

print("Specific Boundary S:", S)

print("General Boundary G:")
for i, g in enumerate(G, 1):
    print("G", i, ":", g)
```

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OUTPUT:



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